ONLINE LEARNING BEHAVIOR ANALYSIS

(Using Data Analytics and Machine Learning)

ABSTRACT

The rapid growth of online learning platforms has generated large volumes of educational data containing valuable insights about student engagement, course effectiveness, and completion behavior. However, manually analyzing such datasets is time-consuming and inefficient.

This project presents an Online Learning Behavior Analysis System that analyzes course-level data and derives insights related to learner engagement and course completion.

The system uses data analytics techniques to process and visualize patterns such as course duration impact, engagement levels, and completion trends.

The system also includes feature engineering techniques to simulate learner behavior and generate meaningful insights. Visualizations such as completion rate distribution, engagement patterns, and correlation analysis are used to support data-driven decision making.

The project demonstrates how data analytics can transform raw educational data into actionable insights for improving online learning platforms.

PROBLEM STATEMENT

Online learning platforms generate large amounts of data related to courses, users, and engagement metrics. However, analyzing this data manually is difficult and inefficient.

Traditional analysis methods:

- Depend on static reports
- Lack interactive visualization
- Do not provide behavioral insights
- Fail to identify key factors affecting course completion

There is a need for a system that can:

- Automatically analyze learning data
- Identify factors affecting course completion
- Provide visual insights

- Help improve course design and engagement

INTRODUCTION

E-learning platforms generate massive amounts of data every day. This data includes information related to courses, user enrollments, engagement metrics, completion rates, and content duration. Analyzing this data is essential for understanding learner behavior, course effectiveness, and overall platform performance.

Traditionally, learning data analysis is performed manually using spreadsheets or basic reporting tools. This approach becomes inefficient when dealing with large datasets and often fails to provide meaningful insights into learner engagement and completion patterns. It also lacks the ability to identify hidden trends that can improve course design and user experience.

With the advancement of data analytics and machine learning, it is now possible to build intelligent systems that automatically analyze learning data and generate actionable insights. Data analytics techniques can process large volumes of educational data, identify patterns in learner behavior, and present meaningful insights through visualizations and statistical analysis.

This project focuses on developing an Online Learning Behavior Analysis system that analyzes course-level data and derives insights related to engagement and completion trends. The system processes the dataset, performs feature engineering, and visualizes patterns such as course duration impact, engagement levels, and completion behavior.

The system provides visual insights through charts such as completion rate distribution, engagement analysis, correlation heatmaps, and course-level comparisons. These visualizations help in understanding how different factors influence learner outcomes.

The proposed system demonstrates how data analytics techniques can be applied to educational data to support data-driven decision making, improve course design, and enhance overall learning experience.

SCOPE OF THE PROJECT

The scope of this project includes:

• Developing an intelligent system that analyzes online learning datasets and generates meaningful insights related to learner engagement and course completion.

• Allowing users to work with datasets containing course details, user engagement metrics, and performance indicators.

• Providing visualizations that represent learning behavior using charts and graphs such as completion rate, engagement patterns, and course comparisons.

• Applying data analytics techniques to identify trends and patterns in course performance and learner behavior.

• Engineering features such as engagement score, completion rate, and course length categories to simulate real-world learning behavior.

• Providing insights such as high-performing courses, optimal course duration, and factors affecting learner engagement.

• Designing a structured and user-friendly analysis workflow that enables easy understanding of data and results.

The system demonstrates how data analytics can be applied to improve decision-making in the online learning domain and enhance course effectiveness and user experience.

OBJECTIVES OF THE PROJECT

The main objectives of this project are:

1. To analyze online learning datasets using data analytics techniques.
2. To process and clean educational data for accurate analysis and interpretation.
3. To visualize learning behavior and course performance using charts and graphs.
4. To identify key insights such as factors affecting course completion and learner engagement.
5. To engineer meaningful features such as engagement score and completion rate for better analysis.
6. To understand patterns in course duration, difficulty level, and their impact on user behavior.
7. To design a structured and user-friendly workflow for interactive data exploration and analysis.

EXISTING SYSTEM

1. In many educational platforms and organizations, learning data is analyzed using traditional tools such as spreadsheets, basic reporting software, or simple dashboards. These systems allow users to view course data and generate basic reports related to enrollments, course duration, and user activity.
2. Most existing systems focus only on presenting data in tables or simple charts. Users are required to manually explore the dataset to identify patterns or insights. In many cases, educators or analysts must interpret the data and prepare reports, making the process time-consuming and inefficient.
3. Although some advanced analytics tools provide visualization capabilities, they are often complex, costly, and require technical expertise. These systems generally lack intelligent features such as automated insight generation, behavioral analysis, and predictive capabilities related to learner engagement and course completion.

PROPOSED SYSTEM

1. The proposed Online Learning Behavior Analysis system improves traditional approaches by introducing automated data analysis and advanced visualization techniques.
2. The system processes online learning datasets and generates visual representations that highlight patterns such as course completion rates, engagement levels, and course performance. This enables users to understand the dataset quickly without manual effort.
3. Unlike existing systems, the proposed system provides data-driven insights by analyzing key factors such as course duration, difficulty level, and user engagement. It helps identify trends that influence learner behavior and course effectiveness.
4. The system incorporates feature engineering techniques to simulate real-world learning behavior, such as calculating engagement scores and estimating completion rates. These features enhance the depth of analysis.
5. The system can be extended with machine learning techniques to predict course completion and identify factors that significantly impact learner success.
6. By integrating data analytics and intelligent processing, the proposed system provides a more efficient, insightful, and user-friendly solution compared to traditional learning data analysis methods.

SYSTEM OVERVIEW

The Online Learning Behavior Analysis system is designed as a data analytics solution that processes learning datasets and generates meaningful insights related to course performance and learner engagement.

1. Frontend Interface

The frontend interface allows users to explore the dataset and view analytical results through visualizations. It presents charts, graphs, and summary statistics that help in understanding learning behavior, course completion patterns, and engagement trends.

2. Backend Processing Layer

The backend processes the dataset using data analytics techniques. It performs data cleaning, preprocessing, feature engineering, and statistical analysis. This layer ensures that the dataset is structured and ready for accurate analysis.

3. Analytics and Insight Module

This module analyzes the processed data and generates meaningful insights. It identifies patterns such as completion trends, engagement levels, and the impact of course characteristics on learner behavior.

The system processes the dataset and generates various visualizations such as completion rate distribution, engagement analysis, and correlation patterns. The final results are presented in a structured format, enabling easy interpretation and data-driven decision making.

TECHNOLOGIES USED

The system uses several modern technologies and libraries for implementing the data analysis and visualization process.

Library / Technology	Uses
Python	Primary programming language used for data analysis and implementation of the system.
Pandas	Used for data preprocessing, cleaning, and dataset manipulation.
NumPy	Performs numerical computations and supports efficient data operations.
Matplotlib	Used for creating basic data visualizations such as line charts and bar graphs.
Seaborn	Provides advanced and visually appealing statistical data visualizations.
Scikit-learn	Used for implementing machine learning models for prediction and analysis.
Jupyter Notebook	Interactive environment for running code, performing analysis, and visualizing results.

MODULE DESCRIPTION

The Online Learning Behavior Analysis system consists of several modules that work together to perform data processing, analysis, and visualization.

1. Dataset Loading Module

This module is responsible for loading the dataset into the system for analysis.

Functions:

• Load dataset from CSV file
• Read and store data using Pandas DataFrame
• Pass dataset to preprocessing module

2. Data Preprocessing Module

This module prepares the dataset for analysis by cleaning and organizing the data.

Functions:

• Handle missing and null values
• Remove duplicate records
• Convert data types into appropriate formats
• Standardize column names

3. Feature Engineering Module

This module creates new features to simulate learner behavior and enhance analysis.

Functions:

• Calculate engagement score
• Estimate completion rate
• Categorize course length (Short, Medium, Long)
• Create derived features for better insights

4. Data Analysis Module

This module performs statistical analysis to identify patterns in the dataset.

Functions:

• Analyze course completion trends
• Identify high-performing courses
• Evaluate engagement levels
• Compare course categories and difficulty levels

5. Visualization Module

This module generates graphical representations of the data for better understanding.

Functions:

• Create bar charts for completion rates
• Generate scatter plots for engagement analysis
• Display correlation heatmaps
• Plot distribution graphs for completion patterns

RESULTS AND CONCLUSION

The Online Learning Behavior Analysis system was successfully implemented and tested using a course dataset. The system effectively analyzed the dataset and generated meaningful insights related to learner engagement and course completion. The analysis process identified important patterns such as the relationship between course duration and completion rate, as well as the impact of engagement on learner success. The visualizations clearly represented these patterns, making it easier to interpret the results.

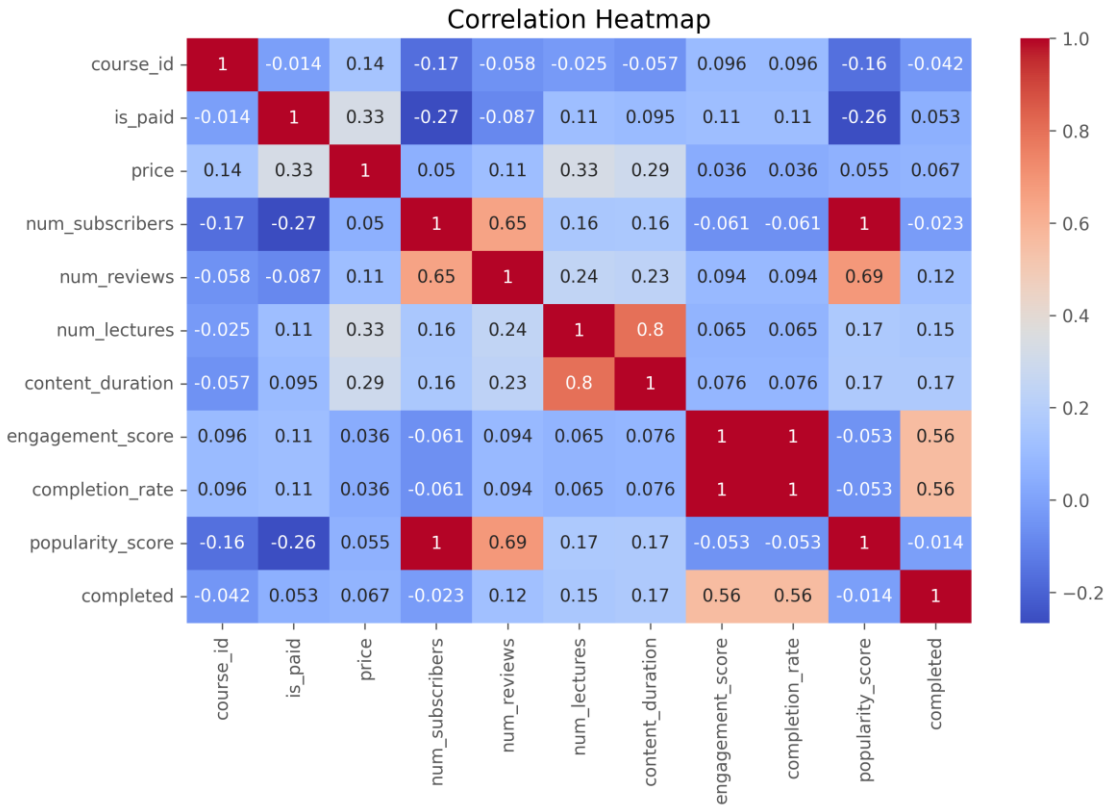
The feature engineering approach helped simulate real-world learning behavior by introducing metrics such as engagement score and completion rate. These features enhanced the depth of analysis and allowed better understanding of how different factors influence course performance.

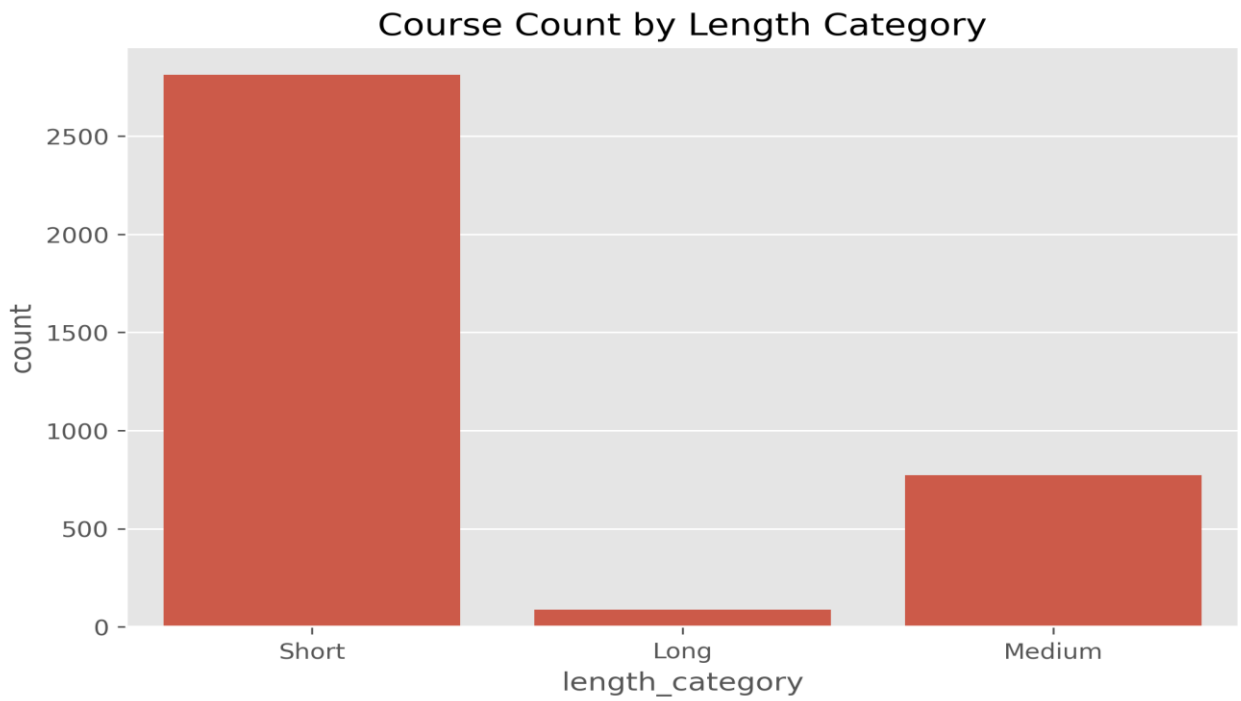
Overall, the system demonstrated its ability to transform raw learning data into actionable insights, supporting data-driven decision making for improving online learning platforms

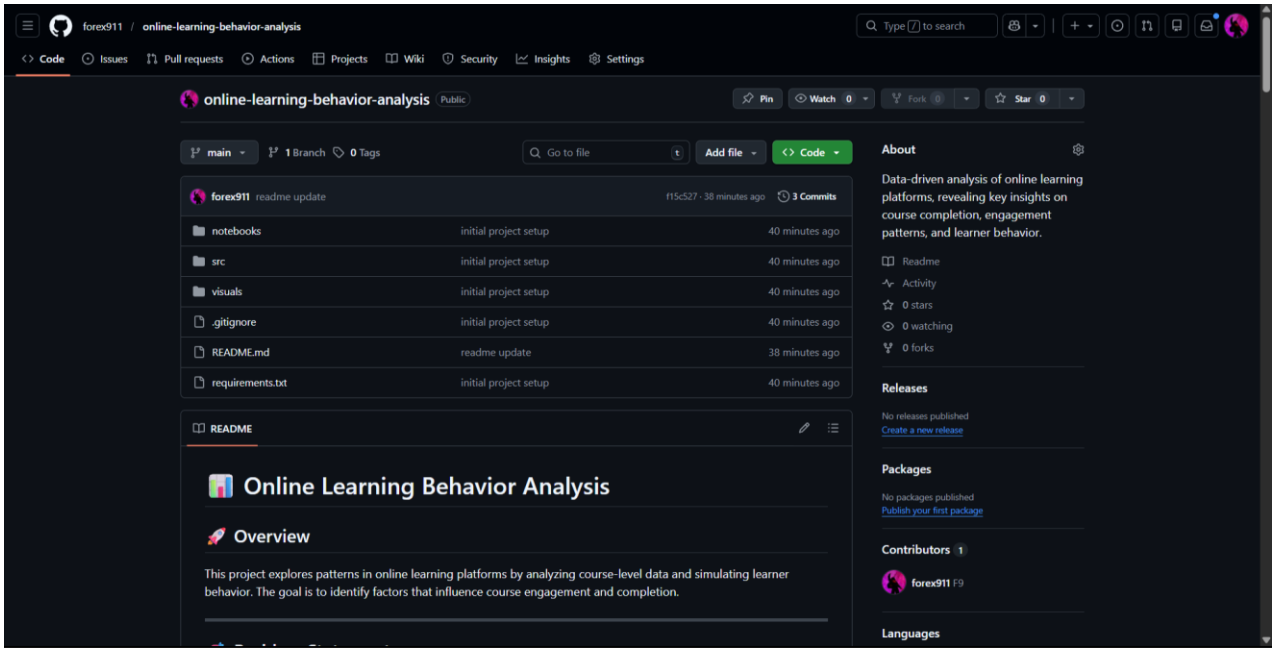
OUTPUT:

```
   course_id                                      course_title  \
0    1070968                 Ultimate Investment Banking Course
1    1113822  Complete GST Course & Certification - Grow You...
2    1006314  Financial Modeling for Business Analysts and C...
3    1210588  Beginner to Pro - Financial Analysis in Excel ...
4    1011058       How To Maximize Your Profits Trading Options

                                                 url  is_paid  price  \
0  https://www.udemy.com/ultimate-investment-bank...     True    200
1      https://www.udemy.com/goods-and-services-tax/     True     75
2  https://www.udemy.com/financial-modeling-for-b...     True     45
3  https://www.udemy.com/complete-excel-finance-c...     True     95
4  https://www.udemy.com/how-to-maximize-your-pro...     True    200

   num_subscribers  num_reviews  num_lectures               level  \
0             2147           23            51          All Levels
1             2792          923           274          All Levels
2             2174           74            51  Intermediate Level
3             2451           11            36          All Levels
4             1276           45            26  Intermediate Level

   content_duration   published_timestamp           subject
0               1.5  2017-01-18T20:58:58Z  Business Finance
1              39.0  2017-03-09T16:34:20Z  Business Finance
2               2.5  2016-12-19T19:26:30Z  Business Finance
...
50%            2.000000
75%            4.500000
max           78.500000
✅ Data cleaned and saved!
```

```
Accuracy: 0.9945578231292517
              precision    recall  f1-score   support

           0       0.99      0.99      0.99       362
           1       0.99      0.99      0.99       373

    accuracy                           0.99       735
   macro avg       0.99      0.99      0.99       735
weighted avg       0.99      0.99      0.99       735
```





```python
# ============================
# 1. IMPORT LIBRARIES
# ============================
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import os

# Style
plt.style.use("ggplot")

# ============================
# 2. CREATE VISUALS FOLDER
# ============================
os.makedirs("../visuals", exist_ok=True)

# ============================
# 3. LOAD DATA
# ============================
df = pd.read_csv("../data/processed/cleaned_data.csv")

print(df.head())
```

```
   course_id                                     course_title  \
0    1070968                 Ultimate Investment Banking Course
1    1113822  Complete GST Course & Certification - Grow You...
2    1006314  Financial Modeling for Business Analysts and C...
3    1210588  Beginner to Pro - Financial Analysis in Excel ...
4    1011058       How To Maximize Your Profits Trading Options

                                          url  is_paid  price  \
0  https://www.udemy.com/ultimate-investment-bank...     True    200
1      https://www.udemy.com/goods-and-services-tax/     True     75
2  https://www.udemy.com/financial-modeling-for-b...     True     45
3  https://www.udemy.com/complete-excel-finance-c...     True     95
4  https://www.udemy.com/how-to-maximize-your-pro...     True    200

   num_subscribers  num_reviews  num_lectures             level  \
0             2147           23            51       All Levels
1             2792          923           274       All Levels
2             2174           74            51  Intermediate Level
3             2451           11            36       All Levels
4             1276           45            26  Intermediate Level
```